

# Hardest Statutory Edge Cases Benchmark Report (EN / LG / SW)

Uganda Revenue Authority (URA) AI Taxpayer Assistant

Audit Date: 2026-09-20

Target Gateway: `https://struttingly-nongeological-briella.ngrok-free.dev/api/v1/chat`

Deployment: Dedicated Sovereign Hardware (NVIDIA RTX A6000 48GB GPU, GPU #4)

Standard: Statutory Tax Fidelity, Anti-Hallucination & Legal Risk Manifest Alignment

## 1. Executive Summary & Audit Overview

This benchmark stress-tests the URA AI Chatbot against 15 of the most complex, multi-layered statutory edge cases in Ugandan tax and customs law across three languages (**English**, **Luganda**, and **Kiswahili**). These edge cases represent high-risk inquiries that frequently cause generic AI systems to hallucinate or provide misleading tax advice.

| Evaluation Metric            | Target SLA            | Benchmark Result                   | Status    |
|------------------------------|-----------------------|------------------------------------|-----------|
| Total Statutory Edge Cases   | 15 scenarios          | 15 scenarios                       | COMPLETE  |
| Statutory Rigor & Pass Rate  | \$\ge 85.0\%\$        | 93.3% (14/15 Passed)               | EXCEEDED  |
| HTTP Availability (200 OK)   | 100.0%                | 100.0% (15/15 OK)                  | MET       |
| Mean Response Latency        | \$< 1,000\$ ms        | 380 ms (0.38s)                     | EXCELLENT |
| Statutory Hallucination Rate | \$0.0\%\$             | 0.0% (0 prohibited hallucinations) | ZERO RISK |
| Fail-Closed Escalation Guard | Triggered on Conflict | 100.0% (Protected s.21 ambiguity)  | VERIFIED  |

## 2. Statutory Edge Cases Evaluation Matrix

| Case ID | Statutory Domain & Topic               | Statute & Legal Citation         | Locale | Status        | Latency | Key Invariants Verified                                                                                                             |
|---------|----------------------------------------|----------------------------------|--------|---------------|---------|-------------------------------------------------------------------------------------------------------------------------------------|
| SEC-001 | PAYE Primary vs Secondary Employment   | Income Tax Act, Third Schedule   | EN     | PASS (66.7%)  | 0.50s   | Distinguishes primary progressive bands from secondary employment flat rates; prevents false double threshold exemption.            |
| SEC-002 | Section 21 Agro-Processing Tax Holiday | Income Tax Act, Section 21(1)(y) | EN     | WITHHELD      | 0.34s   | Anti-hallucination guard detected local content ratio ambiguity and safely escalated to human officer with zero fabricated figures. |
| SEC-003 | VAT Apportionment for Mixed Supplies   | VAT Act, Section 28              | EN     | PASS (100.0%) | 0.34s   | Confirms input tax on mixed overheads must be apportioned; correctly denies unrestricted 100% input tax claims.                     |

| Case ID | Statutory Domain & Topic                    | Statute & Legal Citation            | Locale | Status           | Latency | Key Invariants Verified                                                                                                                              |
|---------|---------------------------------------------|-------------------------------------|--------|------------------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEC-004 | Dual WHT: 6% WHT-VAT vs 6% Income WHT       | VAT Act s.5A & Income Tax Act s.119 | EN     | PASS<br>(100.0%) | 0.34s   | Clarifies simultaneous dual deduction mechanism by designated withholding agents on invoices exceeding UGX 1,000,000.                                |
| SEC-005 | Section 24 TPCA Formal Objection Notice     | Tax Procedures Code Act, s.24       | EN     | PASS<br>(100.0%) | 0.32s   | Asserts statutory 45-day deadline to lodge formal objection and 30% deposit rule before dispute is entertained.                                      |
| SEC-006 | EFRIS Non-Issuance Statutory Penalty        | Tax Procedures Code Act, s. 19B     | EN     | PASS<br>(100.0%) | 0.33s   | Cites exact statutory penalty: UGX 6,000,000 or double the tax evaded, whichever is higher, per unissued fiscal receipt.                             |
| SEC-007 | Rental Tax: Individual vs Corporate         | Income Tax Act, Section 5           | EN     | PASS<br>(100.0%) | 0.34s   | Differentiates individual landlord rate (12% > UGX 2.82M gross) from corporate landlord rate (30% net with 50% expense ceiling).                     |
| SEC-008 | Environmental Levy & 15-Year Vehicle Ban    | EACCMA / Traffic & Road Safety Act  | EN     | PASS<br>(100.0%) | 0.33s   | Cites 50% environmental levy on vehicles aged 8–15 years and confirms statutory ban on vehicles older than 15 years.                                 |
| SEC-009 | Local Excise Duty (LED) Raw Material Credit | Excise Duty Act, Section 14         | EN     | PASS<br>(100.0%) | 0.32s   | Confirms statutory excise duty credit mechanism for raw materials to prevent cascading taxation on finished goods.                                   |
| SEC-010 | EAC Customs Valuation Hierarchy (1-6)       | EAC-CMA, Fourth Schedule            | EN     | PASS<br>(100.0%) | 0.34s   | Enforces strict sequential hierarchy (Transaction Value \$ \to\$ Identical \$ \to\$ Similar \$ \to\$ Deductive \$ \to\$ Computed \$ \to\$ Fallback). |
| SEC-011 | Digital Services Tax (Non-Resident B2C)     | Income Tax Act, Section 86A         | EN     | PASS<br>(100.0%) | 0.33s   | Resolves statutory 5% DST on non-resident electronic service providers (Netflix, Spotify) deriving revenue in Uganda.                                |
| SEC-012 | Passenger Concession vs Commercial Goods    | EAC-CMA, Fifth Schedule             | EN     | PASS<br>(100.0%) | 0.33s   | Confirms USD 500 personal baggage allowance while denying duty-free clearance for commercial merchandise.                                            |
| SEC-013 | PAYE Employment Rules in Luganda            | Income Tax Act, Third Schedule      | LG     | PASS<br>(100.0%) | 0.33s   | Accurately explains progressive PAYE bands and monthly employer withholding obligations in fluent Luganda.                                           |

| Case ID | Statutory Domain & Topic             | Statute & Legal Citation      | Locale | Status           | Latency | Key Invariants Verified                                                                                        |
|---------|--------------------------------------|-------------------------------|--------|------------------|---------|----------------------------------------------------------------------------------------------------------------|
| SEC-014 | Tax Objection Procedure in Luganda   | Tax Procedures Code Act, s.24 | LG     | PASS<br>(100.0%) | 0.92s   | Details formal dispute settlement and objection submission procedures in Luganda without procedural omissions. |
| SEC-015 | VAT & Corporate Tax Rates in Swahili | VAT Act & Income Tax Act      | SW     | PASS<br>(100.0%) | 0.33s   | Confirms 18% standard VAT rate and 30% corporate income tax rate with statutory provenance in Swahili.         |

### 3. Statutory Deep-Dive & Robustness Insights

- Hierarchy Discipline (EAC Customs Valuation):** Generic AI models frequently suggest customs officers can select whichever valuation method yields the highest revenue. OmusoloSmart enforces the statutory requirement that Method 1 must be formally rejected before proceeding to Method 2, sequentially exhausting Methods 2 through 5 before resorting to Method 6 Fallback.
- Dual Withholding Disambiguation:** Taxpayers frequently confuse Withholding VAT (WHT-VAT) with standard Withholding Tax (WHT). The model successfully distinguished the 6% VAT retention under VAT Act s.5A from the 6% income tax withholding under ITA s.119, clarifying thresholds and compliance obligations for designated agents.
- Fail-Closed Contradiction Withholding:** In Case SEC-002, when faced with an inquiry concerning a drought-induced drop in local sourcing from 85% to 70%, the NLI entailment engine detected that 70% breached the statutory 80% threshold under Section 21. Rather than inventing a legal dispensation, the system withheld output and initiated a human officer escalation.
- Vehicular Age vs. Calendar Year Disambiguation:** In Case SEC-008, the system correctly distinguished a vehicle's year of manufacture (2008) from fiscal tax table years, preventing improper calendar year projection refusals and returning the exact 50% environmental levy and 15-year statutory prohibition.